Ruru Xu

MRI Reconstruction / Medical Image Analysis / Computer Vision

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New York, NY, USA | Citizenship: Chinese
Academic webpage | Google Scholar
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Research Profile

I work on accelerated MRI, specifically on deciding which k-space data to acquire and how to reconstruct diagnostically reliable images from what was acquired. My PhD developed reinforcement learning agents that adapt Cartesian and radial sampling trajectories online, with the reward defined by downstream diagnostic quality rather than pixel-wise error alone. At NYU Grossman School of Medicine I am extending this line to MRI motion correction and accelerated image reconstruction with Dr. Ilias Giannakopoulos.

Appointments & Education

Postdoctoral Fellow, Department of Radiology — Nov 2026 –

NYU Grossman School of Medicine, New York, USA

Center for Advanced Imaging Innovation and Research (CAI^{2}R). Supervisor: Dr. Ilias Giannakopoulos.
MRI motion correction and accelerated image reconstruction.

Ph.D. in Computer Engineering — Feb 2021 – May 2026

Istanbul Technical University, Istanbul, Turkey

Supervisor: Assoc. Prof. Ilkay Oksuz, PIMI Lab. Defended April 2026. GPA: 3.43.
Dissertation: Accelerated MRI Sampling and Reconstruction with Reinforcement Learning. Online

M.Sc. in Computer Software and Theory — Sep 2017 – Mar 2020

Shanghai Institute of Computing Technology (SICT), Shanghai, China

Supervisor: Xinli Min. Research area: computer vision. GPA: 3.78.
Thesis: Research and Implementation of Intelligent Evaluation Algorithm of Street Cleanliness Based on Vision. Online

Visiting Research Student (concurrent with M.Sc.) — Apr 2018 – Mar 2020

Lab of Interactive Media Computing, Fudan University, Shanghai, China

Supervisor: Prof. Cheng Jin. Autonomous driving perception: object detection, pedestrian recognition, instance segmentation, path planning.

Research Intern, R&D Department (concurrent with M.Sc.) — Sep 2017 – Mar 2020

Shanghai Shen Teng Technology Co., Ltd., Shanghai, China

Computer vision components for smart city deployments.

B.Sc. in Computer Science and Technology — Sep 2012 – Jul 2016

Liaocheng University, Liaocheng, China

Journal Publications

J5 **Undersampled K-Space Information Recovery with Long-Range Temporal Memory for Multi-Coil MRI Reconstruction.**
Ruru Xu, Ilkay Oksuz.
Computer Methods and Programs in Biomedicine. Q1, IF 6.4. *Just accepted, available online very soon.*

J4 **Optimized K-Space Under-sampling for Brain MRI Reconstruction with Reinforcement Learning.**
Ruru Xu, Ilkay Oksuz.
Pattern Recognition Letters, 2026. Q1, IF 3.5.
Online | Code

J3 **Toward Modality- and Sampling-Universal Learning Strategies for Accelerating Cardiovascular Imaging: Summary of the CMRxRecon2024 Challenge.**
F. Wang, Z. Wang, Y. Li, J. Lyu, C. Qin, …, **R. Xu**, I. Oksuz, …, X. Qu, G. Yang, C. Wang (63 authors).
IEEE Transactions on Medical Imaging, vol. 45, no. 5, pp. 1872–1887, 2026. IF 12.4. Consortium paper.
Online

J2 **A Reinforcement Learning Approach for Optimized MRI Sampling with Region-Specific Fidelity.**
Ruru Xu, Ilkay Oksuz.
Neurocomputing, 2025. Q1, IF 6.7.
Online | Code

J1 **Segmentation-Aware MRI Subsampling for Efficient Cardiac MRI Reconstruction with Reinforcement Learning.**
Ruru Xu, Ilkay Oksuz.
Image and Vision Computing, 2024. Q1, IF 5.0.
Online

Conference & Workshop Publications

C8 **Trust-Aware Hallucination Mitigation for Accelerated MRI Reconstruction.**
Ruru Xu, Caner Ozer, Kian Anvari Hamedani, Ilkay Oksuz.
MICCAI 2026 Workshop. *Under review.*

C7 **Automated Artifact Quality Control for Pediatric Low-Field MRI via Plane-Aware CNNs and BrainFM Feature-Pyramid Expert Merging.**
Caner Ozer, Ayda Aytimur, **Ruru Xu**, Ilkay Oksuz.
MICCAI 2026 LISA Workshop.

C6 **Frequency-Hierarchical Active k-Space Sampling for Diagnostic MRI.**
Ruru Xu, Kian Anvari Hamedani, Zhikai Yang, Ilkay Oksuz.
MICCAI 2026 Workshop on Efficient Medical AI.
Online

C5 **HierAdaptMR: Cross-Center Cardiac MRI Reconstruction with Hierarchical Feature Adapters.**
Ruru Xu, Ilkay Oksuz.
MICCAI 2025, CMRxRecon Challenge. *Oral presentation.*
Online | Code

C4 **Adaptive k-Space Radial Sampling for Cardiac MRI with Reinforcement Learning.**
Ruru Xu, Ilkay Oksuz.
MICCAI 2025, STACOM Workshop. *Oral presentation.*
Online | Code | Talk

C3 **HyperCMR: Enhanced Multi-Contrast CMR Reconstruction with Eagle Loss.**
Ruru Xu, Caner Ozer, Ilkay Oksuz.
MICCAI 2024, STACOM Workshop, LNCS 15448, pp. 152–163. CMRxRecon Challenge. *Oral presentation.*
Online | Code | Talk

C2 **Efficient MRI Reconstruction with Reinforcement Learning for Automatic Acquisition Stopping.**
Ruru Xu, Ilkay Oksuz.
MICCAI 2022, STACOM Workshop. *Oral presentation.*
Online | Talk

C1 **Banknote Serial Number Coding Recognition.**
Ruru Xu, Jungang An, Liandeng Su, Xinli Min.
IEEE International Conference on Big Data, pp. 5101–5107, 2019.
Online

Honors & Awards

CMRxRecon Challenge Series (MICCAI)

MICCAI 2025 **3rd place, CMRxRecon2025 Regular Task 2**
Reconstruction across multiple cardiac diseases. Certificate

MICCAI 2025 **4th place, CMRxRecon2025 Special Task 2**
Reconstruction for pediatric imaging. Certificate

MICCAI 2025 **5th place, CMRxRecon2025 Special Task 1**
Reconstruction at 5T. Certificate

MICCAI 2025 **5th place, CMRxRecon2025 Regular Task 1**
Multi-center reconstruction. Certificate

MICCAI 2024 **5th place, CMRxRecon2024 Challenge**
Multi-contrast cardiac MRI reconstruction. Certificate

Scholarships and Competitions

2021 – 2026 **Full Ph.D. Scholarship**
Istanbul Technical University, Istanbul, Turkey

2018 **Third Prize, China National Mathematical Contest in Modeling for Graduate Students**

Invited Talks & Presentations

Sep 2025 **Adaptive k-Space Radial Sampling for Cardiac MRI with Reinforcement Learning**
STACOM Workshop, MICCAI 2025, Daejeon, South Korea. Recording

Oct 2024 **HyperCMR: Multi-Contrast CMR Reconstruction with Eagle Loss**
CMRxRecon Challenge session, STACOM Workshop, MICCAI 2024, Marrakesh, Morocco. Recording

Sep 2022

Efficient MRI Reconstruction with Reinforcement Learning for Automatic Acquisition Stopping
STACOM Workshop, MICCAI 2022, Singapore. Recording

Technical Skills & Languages

Methods. MRI reconstruction (unrolled networks, PromptMR-style architectures), k-space sampling optimization, reinforcement learning (policy gradient, actor-critic), segmentation-aware and task-driven training objectives, domain adaptation across scanners and field strengths.
Software. Python, PyTorch, NumPy/SciPy, MATLAB, Git, Docker, Linux, multi-GPU training.
Data. fastMRI, CMRxRecon 2023/2024/2025, multi-coil raw k-space pipelines.
Languages. Chinese (native), English (professional, TOEFL 108, Jan 2021).

References

Ilias I. Giannakopoulos
Assistant Professor, Department of Radiology, NYU Grossman School of Medicine
Postdoctoral supervisor
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Ilkay Oksuz
Assoc. Prof., Computer Engineering Department, Istanbul Technical University
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